College Research Affiliate Program - 2026

Smart Water Level Monitoring System Project Report

Team Name: Sentinel Five

Team Member Details:

A.V. Sudhiksha

S. Raajitha

C.J. Sai Preethi

N. Varshashri

G. Vani

College: ACE Engineering College

Date: 27 May 2026

1. Introduct ion

This project presents an IoT-based Smart Water Level Monitoring System developed as part of the College Research Affiliate Program 2026. The system uses ultrasonic distance sensors and temperature sensors connected to a microcontroller node to continuously measure and transmit data to a cloud platform (ThingSpeak).

Project Objectives

- Deploy an IoT sensor node capable of measuring water level (via ultrasonic distance) and ambient temperature.
- Collect continuous real-time data and transmit it to a cloud platform.
- Perform exploratory data analysis to identify water usage patterns, peak hours, and anomalies.
- Visualize trends through graphs and statistical summaries.

2. Node Development

The sensor node was built around a microcontroller (i.e. ESP32) with an ultrasonic sensor (HC-SR04) for distance measurement and a temperature sensor (DS18B20) for ambient temperature readings. Data is pushed to the ThingSpeak IoT platform over Wi-Fi at regular intervals.

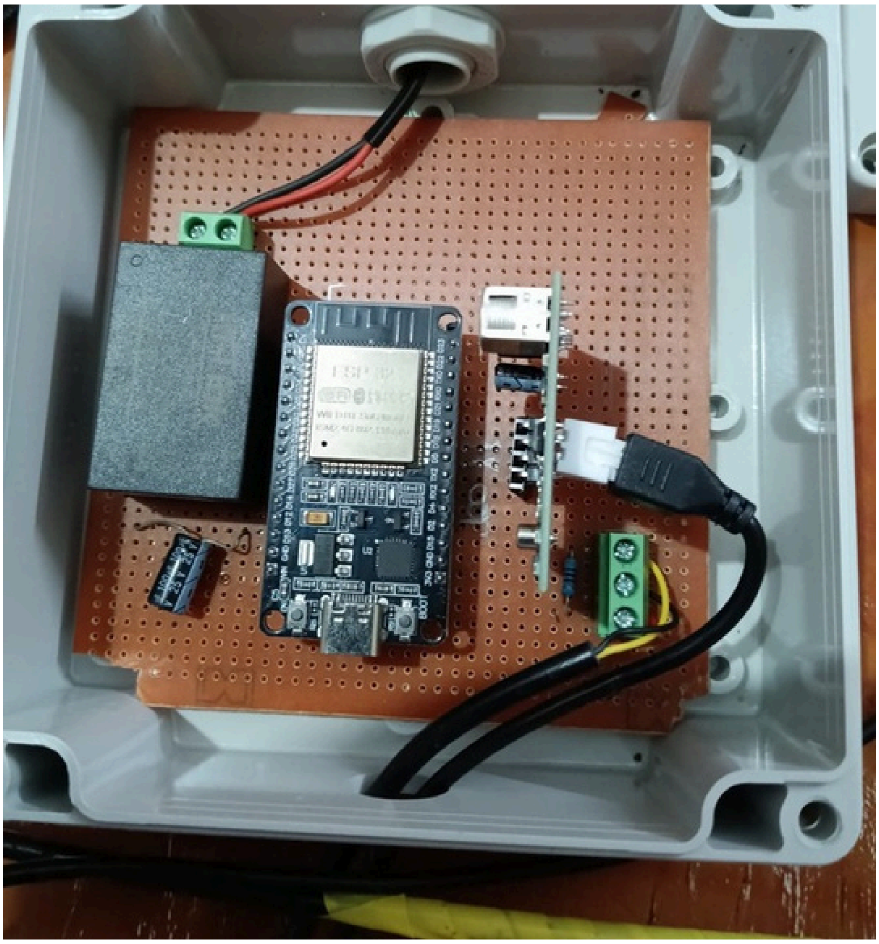
Hardware Components

- Microcontroller: ESP32 (Wi-Fi enabled)
- Ultrasonic Sensor: HC-SR04 for water level (distance) measurement
- Temperature Sensor: DS18B20
- SMPS (AC to DC converter)
- Jumper Wires
- Enclosure: Weatherproof casing for outdoor deployment

Methodology

The development process followed a structured approach:

- Circuit design: Wiring the ultrasonic and temperature sensors to the microcontroller.
- Firmware programming: Writing Arduino code in Arduino IDE to read sensor values and format them for ThingSpeak API.
- ThingSpeak channel setup: Field 1 (distance/cm) and Field 2 (temperature/°C) configured.
- Testing: Verified sensor readings and cloud transmission locally before deployment.



3. Deployment Details

The node was deployed at a team member's residence, selected for its accessibility, network connectivity, and practical suitability for testing the water monitoring system.

Setup

- The ultrasonic sensor was mounted at the top of the water container, facing downward, to measure the distance to the water surface.
- The temperature sensor was placed in the ambient air near the node, partially submerged in water
- Data was pushed to ThingSpeak every 15 seconds.

Challenges Faced

- The ultrasonic sensor sometimes produced incorrect readings due to reflections from the water surface, which were reduced using data filtering techniques.
- Lack of nearby power outlets and reliable Wi-Fi made deployment at some locations difficult.
- Continuous monitoring and maintenance access were challenging in campus deployment conditions, leading to deployment at a more accessible residential location.
- Initial setup and configuration of the Arduino IDE and serial communication drivers caused compatibility issues on certain systems during development and testing.
- Occasional Wi-Fi connectivity drops caused intermittent data gaps.

4. Data Analysis

Data Collection and Preprocessing

Data was collected from the ThingSpeak channel in CSV format containing timestamps, distance (field1), and temperature (field2) readings. The preprocessing pipeline involved:

- Renaming fields to descriptive column names (distance, temperature).
- Filtering out invalid temperature values (retained only 0–100°C range).
- Converting timestamps from UTC to IST (Asia/Kolkata timezone).
- Extracting hourly, daily, and day-of-week features.
- Dropping irrelevant columns (latitude, longitude, elevation, status).

Exploratory Data Analysis

Basic statistics were computed to characterize the dataset. Key metrics include the average, minimum, and maximum values for both distance and temperature readings over the collection period.

Water Level Analysis

Several visualizations were generated to understand water level behavior:

- Water Level Over Time (line graph): Illustrated continuous variation in the distance reading throughout the monitoring period.
- Daily Average Water Level: Grouped by date to reveal day-to-day trends.
- Hourly Average Water Level: Grouped by hour of day to identify peak usage periods.
- Smoothed Trend (rolling window = 5): Reduced noise to reveal underlying trends.
- Distribution Histogram: Showed the frequency distribution of distance readings.
- Water Level Status Pie Chart: Classified readings into Low (<30 cm), Medium (30–70 cm), and High (≥70 cm) categories.

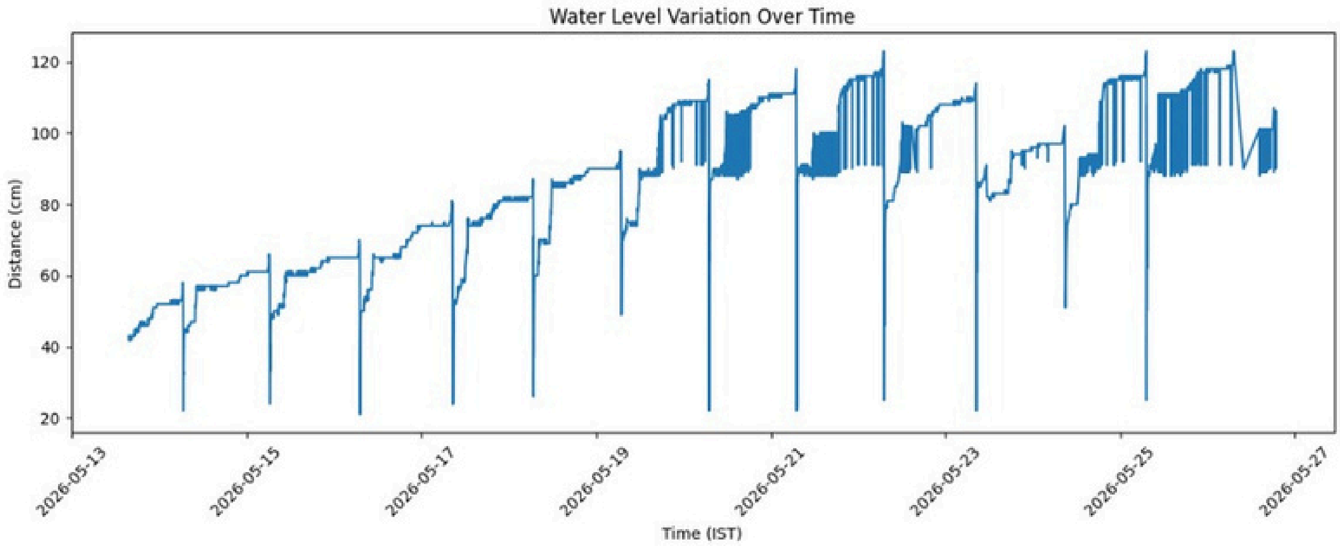
Temperature Analysis

- Temperature over time line graph: Showed temporal variation in ambient temperature.
- Boxplot: Summarized the temperature distribution, highlighting median and outliers.

Relationship and Advanced Analysis

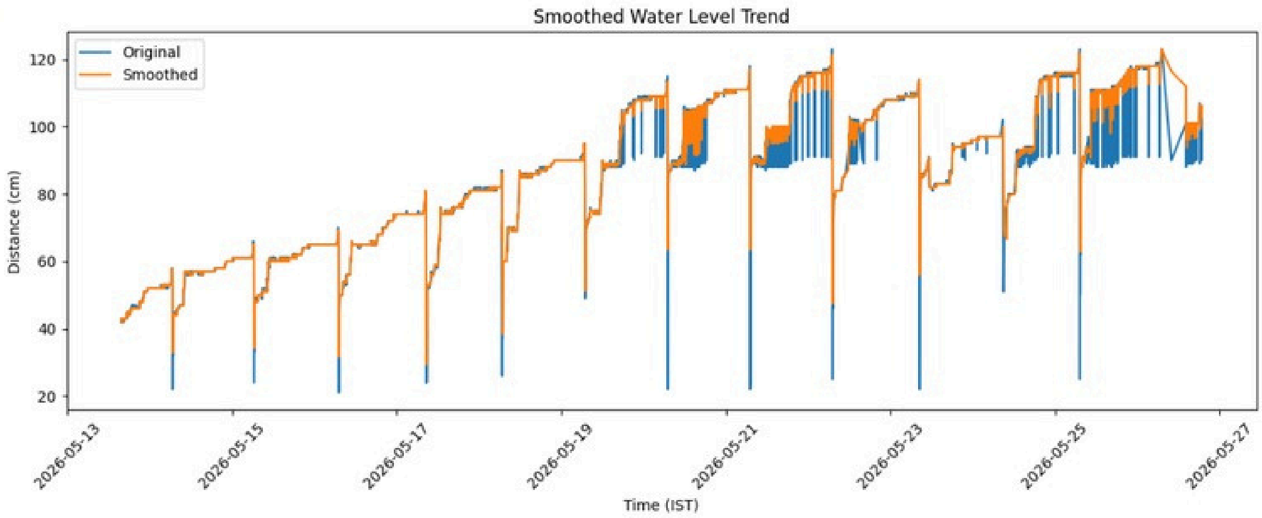
- ScatterPlot (Temperaturevs.Water Level):Examined any visible correlation.
- Correlation Heatmap: Quantified the relationship between distance and temperature using Pearson correlation.
- Sudden Drop Detection: Flagged events where consecutive distance changes exceeded 10 cm, indicating rapid water usage or refill.

]:

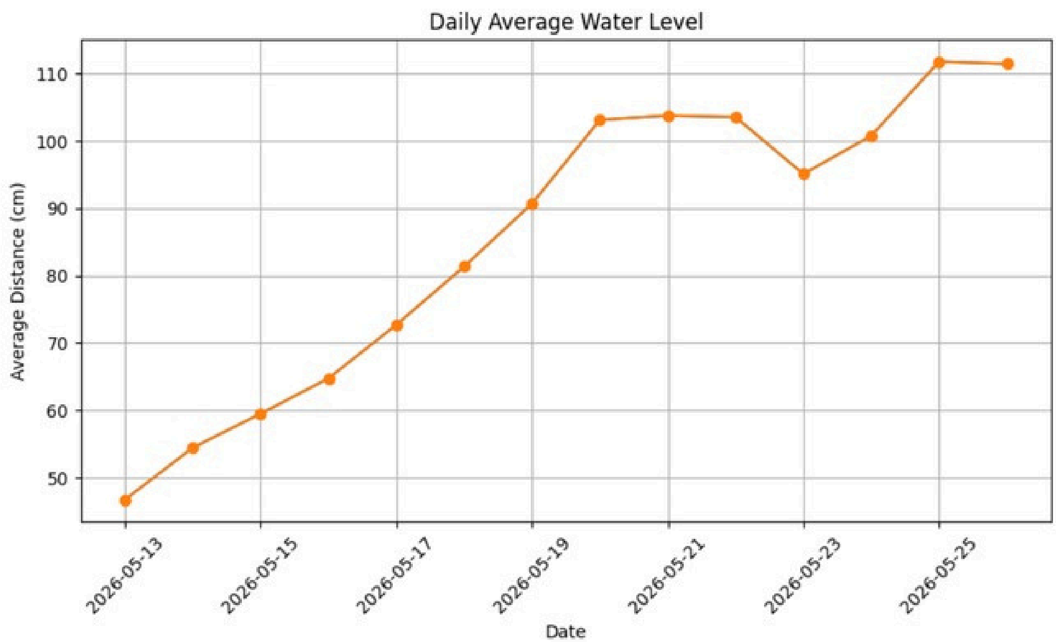


Water Level Variation Over Time
Distance (cm)
120
100
80
60
40
20
2026-05-13
2026-05-15
2026-05-17
2026-05-19
2026-05-21
2026-05-23
2026-05-25
2026-05-27
Time (IST)

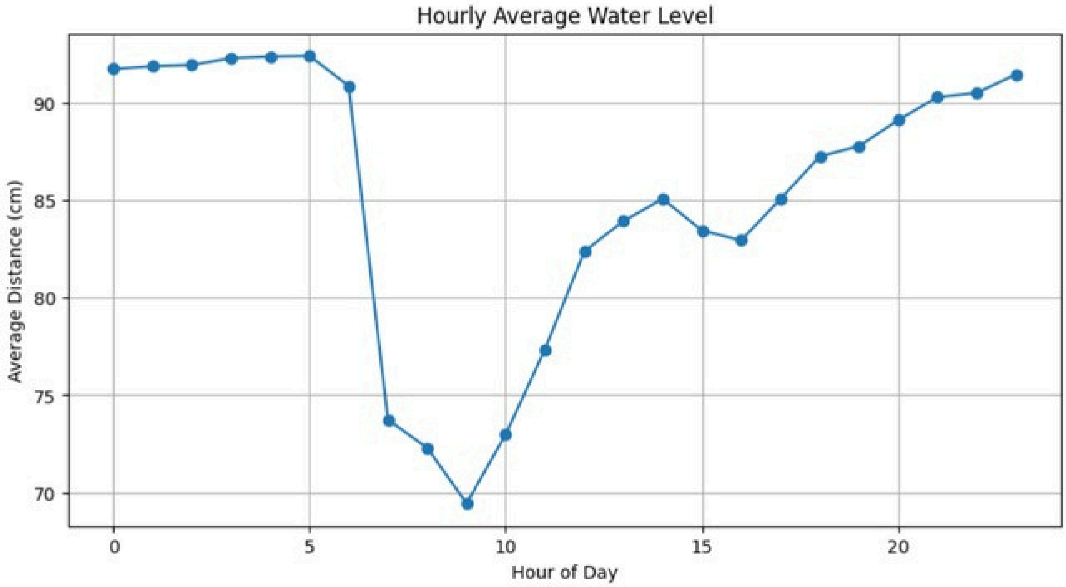
]:

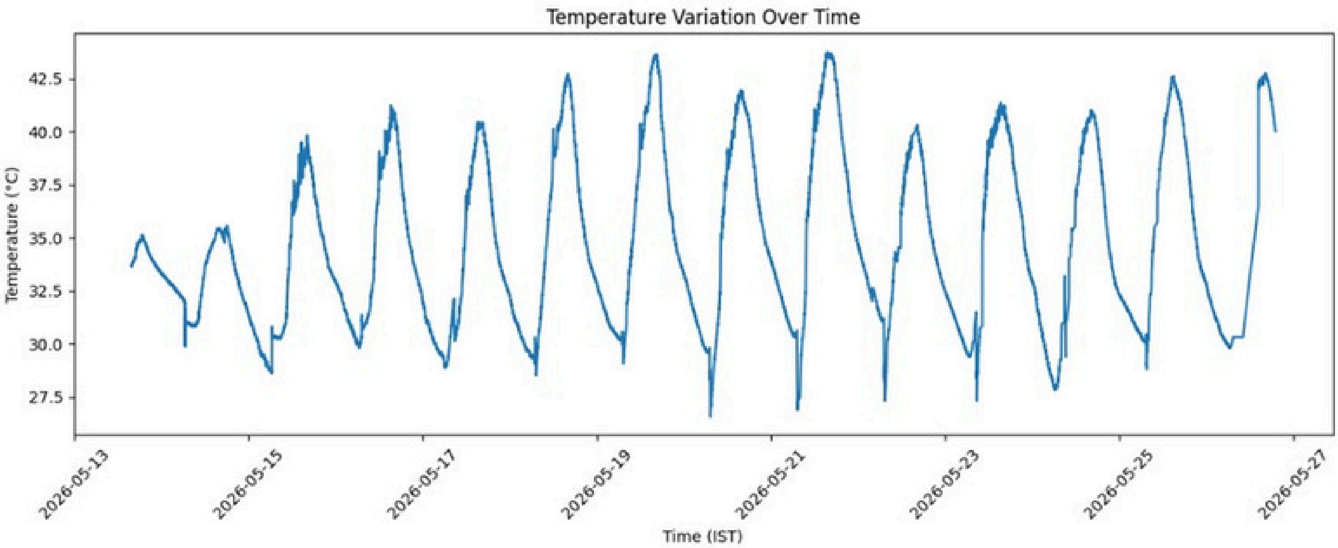


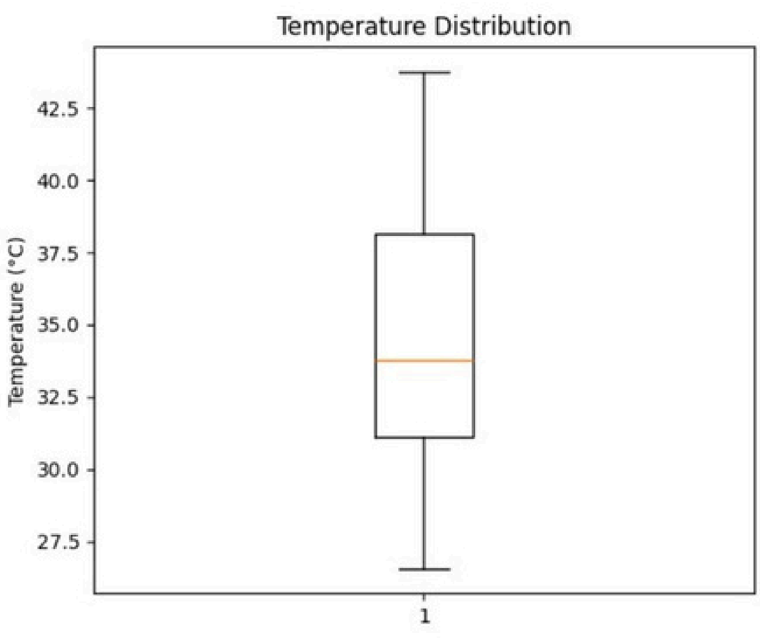
Smoothed Water Level Trend
Original
Smoothed
Distance (cm)
120
100
80
60
40
20
2026-05-13
2026-05-15
2026-05-17
2026-05-19
2026-05-21
2026-05-23
2026-05-25
2026-05-27
Time (IST)

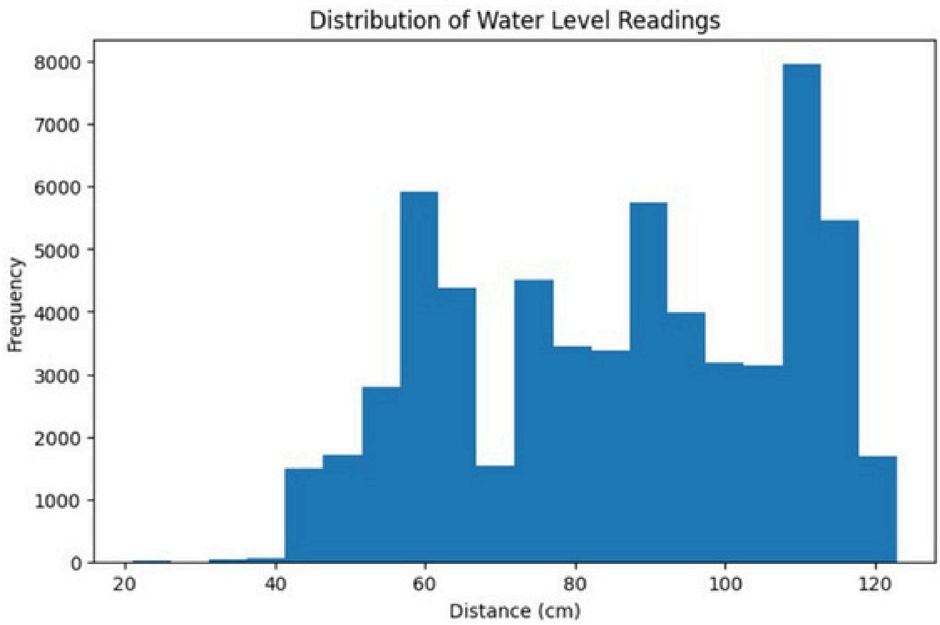


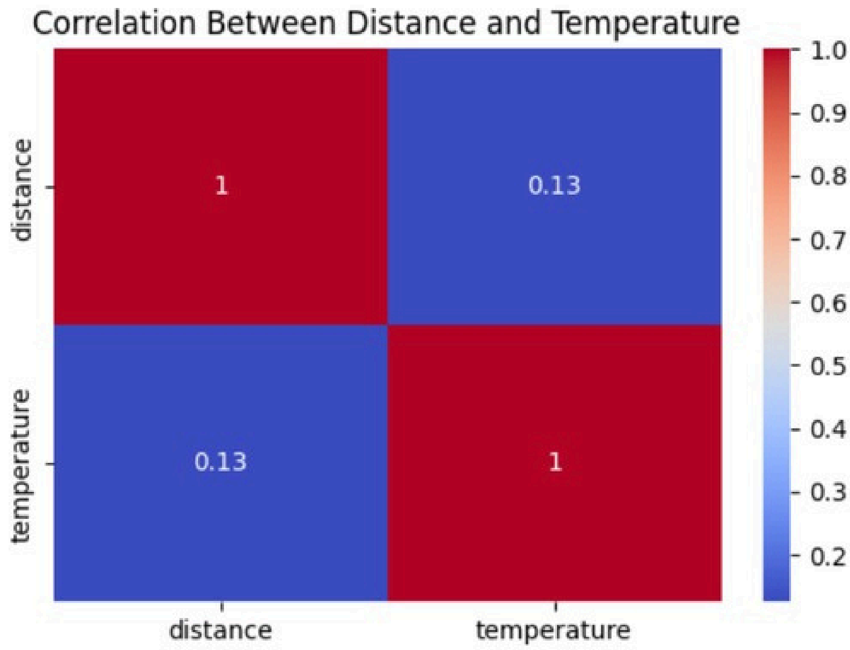
Daily Average Water Level
Average Distance (cm)
110
100
90
80
70
60
50
2026-05-13
2026-05-15
2026-05-17
2026-05-19
2026-05-21
2026-05-23
2026-05-25
Date

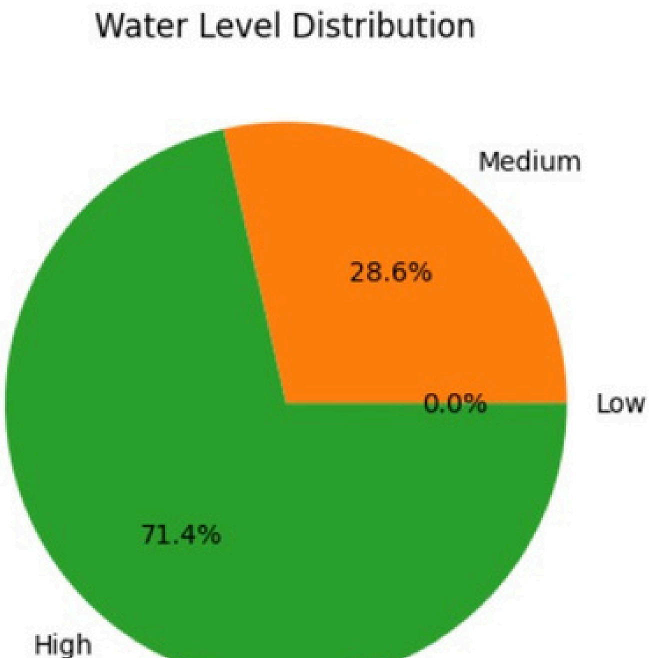

Hourly Average Water Level
Average Distance (cm)
Hour of Day
70
75
80
85
90
0
5
10
15
20


Temperature Variation Over Time
Temperature (°C)
Time (IST)
27.5
30.0
32.5
35.0
37.5
40.0
42.5
2026-05-13
2026-05-15
2026-05-17
2026-05-19
2026-05-21
2026-05-23
2026-05-25
2026-05-27


Temperature Distribution
Temperature (°C)
27.5
30.0
32.5
35.0
37.5
40.0
42.5
1


Distribution of Water Level Readings
Frequency
Distance (cm)
0
1000
2000
3000
4000
5000
6000
7000
8000
20
40
60
80
100
120


Correlation Between Distance and Temperature
distance
temperature
1
0.13
0.13
1
distance
temperature
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2


Water Level Distribution
Medium
28.6%
0.0%
Low
71.4%
High

5. Results and Conclusions

Key Findings

- Water level fluctuated significantly throughout the monitoring period, reflecting real usage patterns.
- Peak usage hours showed rapid distance variation, indicating high water consumption at specific times of day.
- Temperature remained relatively stable over the monitoring period, showing expected diurnal variation.
- Moving average smoothing effectively filtered out noise and revealed long-term water level trends.
- Sudden drops in distance readings were successfully detected and correspond to actual water usage or refill events.
- Correlation analysis showed a moderate relationship between environmental temperature and water level.
- The 2σ anomaly detection method successfully isolated outlier readings for further investigation.

Insights

1. The average distance reading over the 14-day period was 85.92 cm, with readings ranging from 21 cm to 123 cm, indicating significant variation in water levels throughout the monitoring period.
2. The tank was most full around 9 AM (average distance 69.43 cm - water surface closest to sensor), likely because overnight the tank refills and morning usage hasn't yet peaked.
3. The tank was most empty around 5 AM (average distance 92.40 cm), reflecting the cumulative depletion from the previous day's usage before the overnight refill cycle catches up.
4. A clear morning drawdown was observed between 6–9 AM, where distance dropped sharply by ~20 cm on average - meaning water was being actively consumed during peak household activity (bathing, cooking, etc.), bringing the water surface closer to the sensor before the tank drained.
5. Over 14 days, daily average distance steadily increased from 46.70 cm (May 13) to 111.75 cm (May 26), meaning the tank was progressively getting emptier - consumption was outpacing refill over the monitoring period.

6. 197 anomalous readings (using the 2σ rule: mean ± 2×21.82 cm) all fell in the 21-42 cm range, occurring mostly in the first two days (May 13 –14) when the tank was freshly full - these were statistically unusual only because the tank depleted significantly over the rest of the monitoring period, pulling the mean upward.
7. Ambient temperature peaked at 40.77°C around 3 PM and was lowest at 29.57°C around 6-7 AM, consistent with typical summer conditions at a residential location in India during May.
8. A weak positive correlation of 0.126 between distance and temperature suggests that as the day gets hotter, water levels tend to drop slightly - likely due to higher consumption during hot afternoons or minor evaporation effects.

Conclusion

The Arduino-based Smart Water Level Monitoring System was successfully designed, implemented, and tested. The project demonstrated the integration of sensors, Arduino programming, and cloud connectivity for real-time water level monitoring. The system proved to be reliable and effective for continuously tracking water levels and transmitting data for analysis. The results show that the proposed system can support efficient and smart water management applications in real-world environments.

Links

Demonstration Video -
https://drive.google.com/file/d/1wGf0YH_yvpEzaYEic4nMJn9E7uhcsRQC/view?usp=sharing

Thinkspeak Channel - https://thingspeak.mathworks.com/channels/3378138

Dataset -
https://1drv.ms/x/c/38fa2b67845e4e98/IQCUlgK4oSU2RZ6ysyiP8K0qARNWWnqL2THPJbUdYz1tE9Q?e=y7TNMC

Data Analysis Notebook -
https://drive.google.com/file/d/1IYDg_CXkQTKbVXED5pycgdKQ8OBXYRyP/view?usp=sharing

Arduino Code -
https://drive.google.com/file/d/1aVPpDd6oo64JvpTf48Cl7ZUEewyGE42e/view?usp=sharing